

Critical Analysis: Uncertainty Quantification for In-Context Learning of Large Language Models

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1 Methodological Strengths

1.1 Problem formulation that isolates uncertainty sources specific to in-context learning

A key methodological contribution is the explicit framing of in-context learning (ICL) uncertainty as arising from two distinct factors: (i) the choice and quality of few-shot demonstrations in the prompt (Aleatoric Uncertainty, AU), and (ii) model-side configurations and decoding behaviors (Epistemic Uncertainty, EU). This distinction is operationalized with concrete motivating examples showing demonstration-induced errors and configuration-induced variability across decoding strategies.

The paper further provides a probabilistic formulation of ICL as Bayesian inference with a latent concept variable and an approximate predictive distribution. Although explicitly described as a hypothesis, this Bayesian view is used consistently to justify decomposition targets and estimation choices.

1.2 Mutual-information-based decomposition aligns with established uncertainty principles

The decomposition is built from a mutual-information perspective to separate epistemic and aleatoric components in a way that matches classical uncertainty decomposition (total predictive uncertainty equals expected data uncertainty plus information gained about model uncertainty). This design choice is methodologically strong because it gives AU and EU clear semantics rather than treating uncertainty as an undifferentiated score.

1.3 Entropy-based estimation addresses free-form LLM outputs

A major practical barrier in LLM uncertainty quantification is that outputs are free-form text, so a discrete predictive distribution over task labels is not directly available. The paper proposes an entropy-based estimation pipeline that uses beam search outputs and aggregates them into an answer distribution, enabling computation of uncertainty metrics from generation probabilities. This approach is explicitly designed to handle the free-form output format rather than assuming a closed-set classifier head.

1.4 Evaluation covers multiple uncertainty-relevant regimes beyond standard accuracy

The experimental design evaluates uncertainty in several regimes that are directly tied to reliability:

- **Uncertainty estimation under correct versus incorrect predictions** using AUPR and ROC to measure how well uncertainty separates errors from correct outputs across datasets and model sizes.

- **Out-of-domain prediction detection** with explicit AUROC reporting and percent improvement over a baseline uncertainty score, showing practical value for identifying when model outputs should be distrusted.
- **Demonstration-level out-of-distribution analysis** that varies the demonstration pool composition, including relevant demonstrations and out-of-domain demonstrations, to test the sensitivity expected under ICL.
- **Semantic out-of-distribution detection** as an application scenario with explicit comparison to a semantic-uncertainty baseline.

This breadth is a methodological strength because it stress-tests uncertainty measures under distribution shift and prompt shift rather than restricting evaluation to in-distribution classification.

1.5 Controlled prompting and sampling settings are partially specified

The paper documents key inference-time settings used to generate uncertainty estimates, including beam size and maximum new tokens, and specifies multiple demonstration selection strategies (class-based, random, in-domain relevant, in-domain random, and out-of-domain). It also provides dataset-specific numbers of demonstrations per prompt, supporting reproducibility for the few-shot protocol.

2 Key Limitations

2.1 Task scope is restricted to deterministic classification

The paper explicitly focuses on deterministic NLP tasks such as text classification. This scope simplifies the definition of correctness and enables AUPR/ROC evaluation, but it limits conclusions about uncertainty decomposition for open-ended generation, where both correctness and a suitable latent-variable interpretation become more difficult. The limitation is explicitly acknowledged, with future work stated as extending to generation tasks and handling correctness evaluation challenges.

2.2 White-box dependence is fundamental for the core mutual-information pipeline

The proposed mutual-information-based uncertainty quantification relies on access to token probabilities in generated sequences, which is feasible for white-box models but not for black-box LLM APIs. The paper notes this constraint and provides a variance-based decomposition in the appendix, but the main empirical validation centers on settings where token probabilities are available. As a result, practical applicability to black-box deployment scenarios is not established in the reported experiments.

2.3 Modeling assumptions in the Bayesian approximation are not empirically validated

The predictive distribution is approximated using a Bayesian-neural-network-style likelihood with a Gaussian assumption. This approximation provides an interpretable scaffold but is not accompanied by an empirical study that tests sensitivity to the Gaussian assumption, the choice of covariance structure, or alternative generative approximations.

2.4 Limited reporting of computational cost for uncertainty estimation

The estimation procedure uses beam search and repeated sampling across demonstrations and configurations. While key decoding parameters are provided (beam size and max tokens), detailed cost reporting (such as number of model forward passes per test instance, wall-clock runtime, and hardware) is not specified in the paper. This omission makes it difficult to assess feasibility under realistic latency and budget constraints, especially for larger models.

2.5 Evaluation concentrates on a subset of reliability metrics

The experiments emphasize AUPR, ROC, and AUROC for error and shift detection. These metrics are appropriate for ranking-based uncertainty evaluation. However, calibration-focused metrics and decision-theoretic evaluations (such as risk-coverage curves or abstention-utility trade-offs) are not specified in the paper, which limits conclusions about downstream decision-making based on AU and EU.

3 Technical Bottlenecks

3.1 Representing uncertainty over free-form outputs requires lossy discretization

The estimation pipeline constructs an answer distribution by aggregating probabilities from generated sequences into discrete answer candidates. This step faces an inherent bottleneck: multiple textual realizations can correspond to the same semantic label, while small formatting changes can create distinct strings. Any discretization or normalization rule can introduce artifacts in the estimated entropy, creating a tension between linguistic variability and label-level uncertainty.

3.2 Disentangling demonstration-induced and configuration-induced uncertainty remains confounded

ICL couples demonstrations and decoding: the prompt affects the internal state used for decoding, and configuration choices alter which completions are surfaced. The paper operationalizes AU as demonstration-linked variation and EU as configuration-linked variation. However, the underlying causal structure is not directly identifiable from outputs alone, so the decomposition is intrinsically constrained by confounding between prompt composition and decoding dynamics.

3.3 Black-box uncertainty decomposition lacks a validated estimation pathway

The paper states that mutual-information quantification requires token probabilities and is not applicable to black-box LLMs, offering a variance-based alternative in the appendix. A technical bottleneck is the absence of a validated estimator that achieves comparable decomposition fidelity using only sampled text outputs, which is the dominant access pattern in many real deployments.

3.4 Benchmarking uncertainty under distribution shift is definition-dependent

Out-of-domain prediction detection and demonstration out-of-distribution detection depend on how shift is constructed (sampling rules, relevant versus random demonstrations, domain boundaries). These definitions substantially influence measured gains, creating a bottleneck for generalizing conclusions across tasks where domain boundaries are less clean than dataset-level partitions.

4 Research Implications

4.1 Uncertainty in ICL is strongly tied to the prompting process, not only to the base model

The results indicate that uncertainty decomposition provides actionable signals specifically for ICL, where prompt demonstrations are a first-class source of variability. For example, in out-of-domain prediction detection, EU achieves higher AUROC than a baseline uncertainty score under both class sampling and random sampling (0.761 versus 0.725, and 0.746 versus 0.648, respectively), indicating that configuration-linked uncertainty is informative for identifying unreliable predictions under prompt-driven shift.

4.2 Epistemic uncertainty appears especially useful for shift detection, beyond semantic baselines

In semantic out-of-distribution detection, EU substantially improves ROC relative to a semantic uncertainty baseline in multiple settings, such as ROC 0.658 versus 0.532 for a LLaMA-7B setting. This suggests that decomposition can add information beyond clustering-based semantic disagreement, with implications for reliability pipelines that must detect unfamiliar inputs.

4.3 Model size and demonstration sampling strategy affect uncertainty behavior

The paper reports that model performance improves with class-based sampling and larger model size, and notes that token-level baselines can be weakened by redundancy in natural language outputs. A concrete example is a row where EU ROC reaches 0.648 while semantic uncertainty ROC is 0.548 under a LLaMA-13B setting, reflecting that decomposition benefits can persist even when semantic baselines are competitive. These findings imply that uncertainty quantification in ICL is not model-agnostic: prompt construction and decoding protocols must be treated as part of the system, not as minor implementation details.

5 Potential Research Directions

5.1 Extend decomposition to generation tasks with task-grounded correctness definitions

The paper explicitly identifies extension to generation tasks as future work and notes the challenge of evaluating correctness. A concrete direction is to define correctness via task-grounded evaluators (such as reference-based factuality checks for question answering with known answers, or constrained generation tasks with verifiable outputs) and then re-derive AU and EU on semantic units rather than raw strings.

5.2 Develop black-box estimators that do not require token probabilities

A priority direction is to operationalize the variance-based decomposition for black-box access patterns, validating whether AU and EU estimates derived solely from multiple sampled responses correlate with failure and shift detection in the same manner as probability-based estimates.

5.3 Ablate estimation design choices in the entropy-based pipeline

A systematic ablation suite is needed to quantify how beam size, maximum generation length, answer normalization, and aggregation of semantically equivalent outputs influence entropy estimates. The paper specifies beam size and max new tokens, enabling such controlled follow-up studies.

5.4 Connect AU and EU to actionable decision policies

Given that EU improves AUROC for out-of-domain prediction detection (for example, 0.746 versus 0.648 under random sampling), a concrete direction is to build selective prediction or abstention policies that threshold EU and AU separately and evaluate utility under fixed error budgets.

5.5 Improve semantic aggregation for uncertainty estimation

The bottleneck created by string-level diversity motivates a direction that aggregates generations into semantic classes before entropy computation, using task-specific parsing rules or embedding-based clustering. This can be directly evaluated in semantic out-of-distribution detection settings where semantic uncertainty baselines are already used.

6 Conclusion

The paper provides a well-motivated and technically grounded framework for decomposing uncertainty in in-context learning into aleatoric and epistemic components, supported by a mutual-information foundation and an entropy-based estimator for free-form LLM outputs. Empirically, epistemic uncertainty shows clear advantages for detecting out-of-domain predictions, with AUROC improvements over a baseline uncertainty score across sampling strategies (0.761 versus 0.725 under class sampling, and 0.746 versus 0.648 under random sampling). At the same time, the scope is explicitly limited to deterministic classification tasks, and the main pipeline relies on white-box access to token probabilities, leaving black-box deployment and open-ended generation as primary gaps. The most promising research directions are therefore (i) extending decomposition to generation with verifiable correctness criteria, (ii) validating probability-free estimators for black-box LLMs, and (iii) strengthening robustness through ablations and decision-theoretic evaluations.